Exercise 3: Iterating, and working with strings

1. The Fibonacci sequence is a sequence of numbers (beginning with [1, 1]), where the next number in the sequence is the sum of the two which preceded it. Write some code which keeps track of the last two numbers and the next number using variables, and computes the first 20 elements of the Fibonnaci sequence.

2. The "sum of digits" of a number is found by adding all the digits of a number together. For example, the sum of digits of 234 is 2 + 3 + 4 = 9. Write some code to take a numerical value and print out the sum of digits of that number.

3. Given an array of numbers and a number k, write Python code to check whether the sum of any two numbers from the list is equal to k or not. For example, given [1, 5, 11, 5] and k = 16, your program should return "true", since 11 + 5 is 16.

4. Write a function which accepts a String as input, and returns true if that String is a palindrome, false if not (a palindrome is a word that is spelt the same backwards and forwards).